Fundamental and technical signals for A-share stock selection

Investments 1005 | Group 9

Historical daily data: 2 Jan 2018 to 29 Apr 2026. The main implementation is a long-only, equal-weight Top-100 portfolio with targets refreshed every 20 market sessions.

Research question

Can accounting and technical information improve stock selection after trading costs within the same opportunity set? We compare a fundamental score, a combined score, Ridge, Lasso and a shallow LightGBM model. A matched equal-weight portfolio isolates stock-selection performance from differences in the available stock pool.

Main findings

Ridge earns 395.3% net over the complete history, versus 161.5% for the benchmark. Its maximum drawdown is 33.0%, compared with 32.4%. Higher historical return therefore does not imply lower downside risk.

2026 through 29 Apr 2026

Portfolio	2026 gross	2026 net	Net excess (pp)
Fundamental	3.84%	3.53%	-4.25
Combined	10.21%	9.73%	+1.33
Ridge	9.40%	8.80%	+0.56
Lasso	8.19%	7.52%	-0.65
LightGBM	6.08%	5.56%	-2.40
Benchmark	8.23%	8.18%	+0.00

For 2021–2025, Ridge returns 195.7% net, versus 76.6% for the benchmark. In 2026 through 29 Apr 2026, the corresponding returns are 8.80% and 8.18%; cumulative daily net excess is +0.56 percentage points.

Lasso returns 96.10% net in 2021–2025 and 7.52% in 2026, trailing Ridge in both intervals. Its 2021 null model requests cash, so it is not fully invested in every year. Absolute returns compound; excess sums daily strategy-minus-benchmark returns. Partial-year evidence remains limited.

1. Economic rationale and factor definitions

Profitability, growth, valuation and leverage may help explain relative returns. Price trends, volatility and turnover may capture information diffusion, reversal or liquidity. These are economic hypotheses; a plausible explanation alone does not establish a profitable trading rule.

ID	Prepared input and investment hypothesis
F1	Four-report weighted ROE relative to its trailing 60-day mean: changing profitability.
F2	120-day change in the four-report ROA slope / trailing 120-day standard deviation.
F3	The same slope-change transformation applied to revenue YoY growth.
F5	120-day change in debt/equity / trailing 120-day standard deviation: financial risk.
F6	Latest visible parent equity / total market value, retaining negative equity.
F7	Visible parent-profit TTM / total market value, retaining losses.
F8	Latest visible yearly ROA level: profitability.
F9	Latest visible operating cash flow / revenue: cash generation.
F10	Negative latest visible debt/assets: lower leverage.
T1, T2	Adjusted-close returns over 20 and 60 sessions: trend or reversal.
T3	Trailing 20-session population standard deviation of daily returns: volatility.
T5	Trailing 20-session mean turnover: trading activity and liquidity.
T6	Adjusted close at t-20 / adjusted close at t-126 minus one: medium-term momentum.

The adopted annual models use the union F1, F2, F3, F5, T1, T2, T3, T5, F6, F7, F8, F9. The 14 candidates above also define the common complete-input stock pool. F10 and T6 need not enter an adopted forecast to affect that common pool. Factor directions and annual input lists are fixed in config.json; the full formulas, report selection and missing-value conventions are in data_dictionary.md.

Financial inputs require report periods and both recorded announcement dates no later than the signal day. This prevents using known later report dates, but cannot certify that vendor snapshots preserve the original unrevised values.

2. Historical data and the common opportunity set

The prepared daily panel aligns market prices, volume, turnover, historical universe membership, ST status, listing age, industry, market capitalization and financial snapshots on a fixed historical security axis. Inactive and subsequently delisted instruments remain on that axis; membership is not determined by survival today.

Data characteristic	Recorded value
Market sessions / historical slots	2,018 / 6,000
Mean base-eligible names	4,167
Mean common complete-input names	3,688
First signal / first execution	11 Jul 2018 / 12 Jul 2018

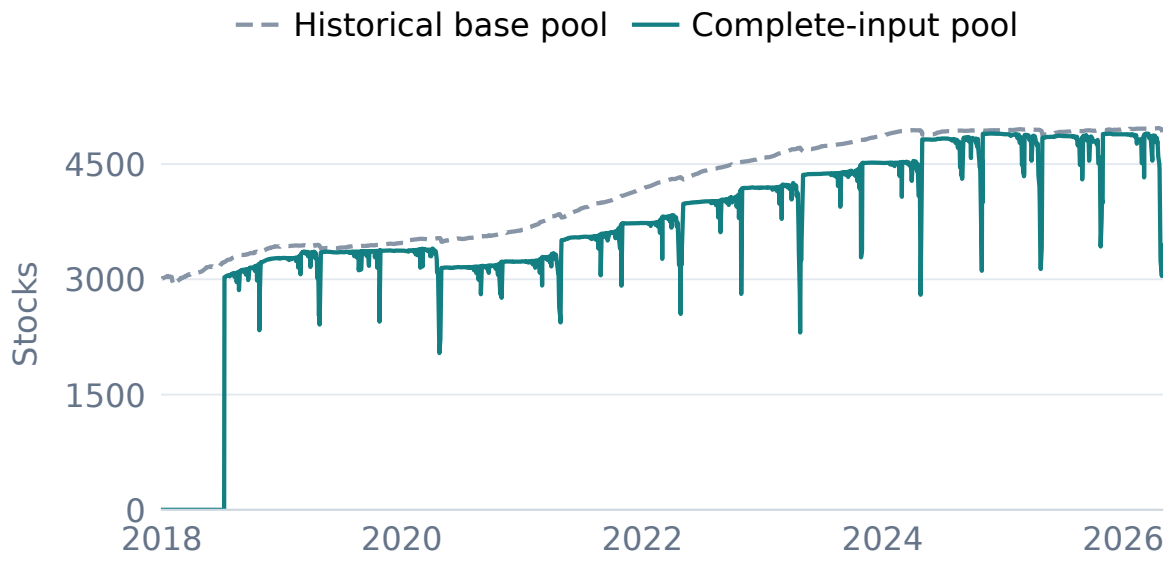


Figure 1. Daily base-eligible and common complete-input stock counts. These are available names, not actual portfolio holdings. Table means include every recorded session and the initial zero complete-input counts.

Eligibility requires historical Shanghai/Shenzhen A-share membership, non-ST status, at least six months listed, positive traded volume and valid market inputs. The common pool additionally requires all 14 candidate values. Every model, raw baseline and benchmark uses this same pool; future-label availability does not determine whether a stock can be selected. The benchmark equal-weights this pool with the same cadence, execution limits and matched costs; it is not the CSI500 price index.

Complete-input filtering favors firms with a longer and more complete reporting history. Initial cash while lagged features accumulate explains the flat beginning before 12 Jul 2018.

3. Models and implementable portfolios

Each daily cross section is converted to centered average ranks. The prediction target is the adjusted opening return from t+1 to t+21, demeaned across eligible labeled stocks.

Method	Construction and fixed settings
Fundamental	Mean signed financial ranks. Score EMA span 20.
Combined	Financial and technical family scores receive equal weight. Score EMA span 40.
Ridge	L2-regularized regression; alpha=10000. [1] Score EMA span 20.
Lasso	L1 regression; historical time-validation selects alpha. [5] Score EMA span 20.
LightGBM	Depth 3; 7 leaves; 56 trees; minimum leaf 500; learning rate 0.05. [2] Score EMA span 40.

Scores are residualized against industry and within-industry log size, reranked, and causally smoothed with EMA alpha=2/(span+1), then reranked again. State continues across years; missing inputs emit no score. Returns and NAV are not smoothed.

Annual refits use expanding prior history sampled every 5 market sessions. A training observation is admitted only when its t+21 exit is strictly earlier than the prediction start. The 2018–2020 portion is a fitted replay using the pre-2021 model; from 2021, annual models use mature earlier labels. Lasso uses the same eight inputs as Ridge; its regularization is selected inside each historical training window (Section 13).

Execution and costs

Top-100 equal-weight targets refresh on one continuous 20-session calendar and trade at the next open using preceding-close scores. Valuation and blocked-order retries remain daily. Suspensions, price limits, cash and T+1 can prevent targets being reached. Gross and net books are simulated separately and compared with benchmarks under their respective fee assumptions.

Commission is 3 bp and slippage 5 bp per side. Sell stamp duty is 10 bp before 28 Aug 2023 and 5 bp thereafter. [3, 4] Round lots, minimum commissions and liquidity impact are omitted.

The causal adjusted-price chain approximates reinvested corporate actions. Sourced exchanges and cash exits follow settlement assumptions; quote-less suspensions retain their last valid mark. Formal unresolved listing ends assume zero recovery rather than proving zero economic value.

4. Complete-history absolute returns

Absolute curves show portfolio NAV minus one, using the complete daily history. They retain cash, execution constraints and compounding along each path.

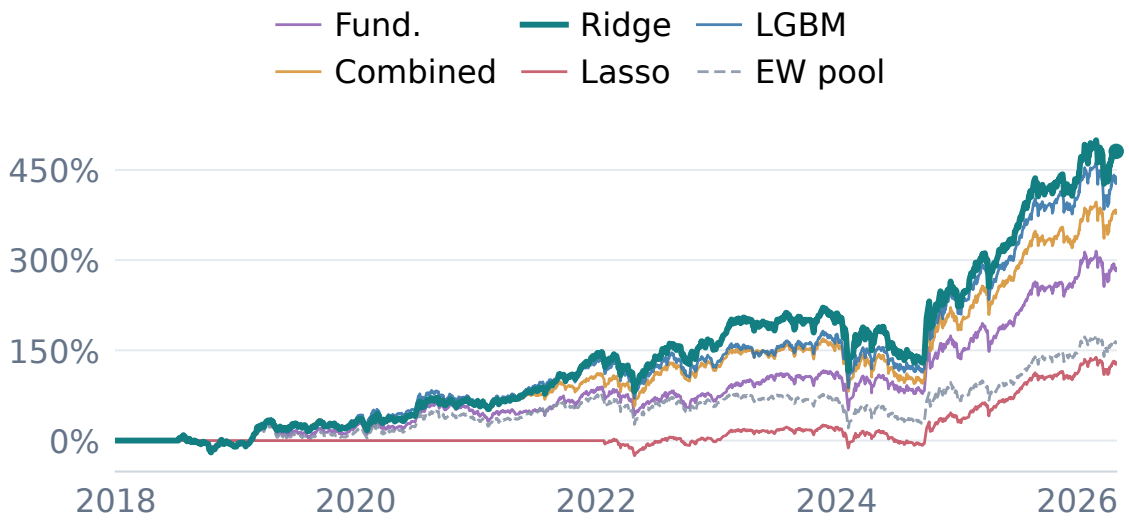


Figure 2. Cumulative gross returns, 2 Jan 2018–29 Apr 2026. Each strategy is compared with the gross common-pool benchmark.

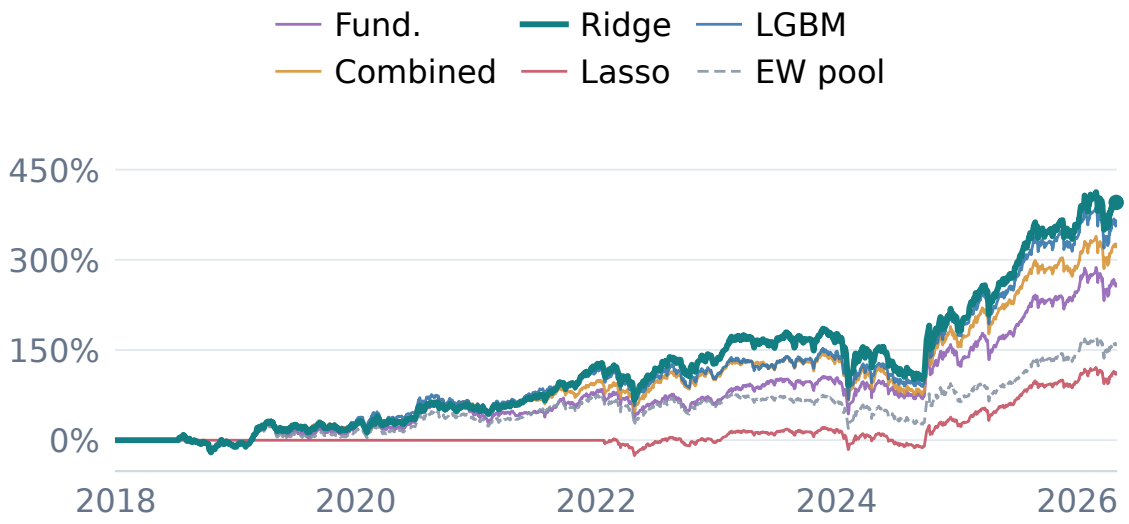


Figure 3. Cumulative net returns over the same complete history, including commission, sell stamp duty and slippage.

Ridge ends at 480.9% gross and 395.3% net. The benchmark ends at 165.5% gross and 161.5% net. Gross-minus-net terminal wealth includes compounding and differing trading paths; it is not the simple sum of fee payments.

5. Complete-history excess returns

Excess sums daily strategy returns minus the matching benchmark, in percentage points. The benchmark is zero by construction. This is not compounded relative wealth.

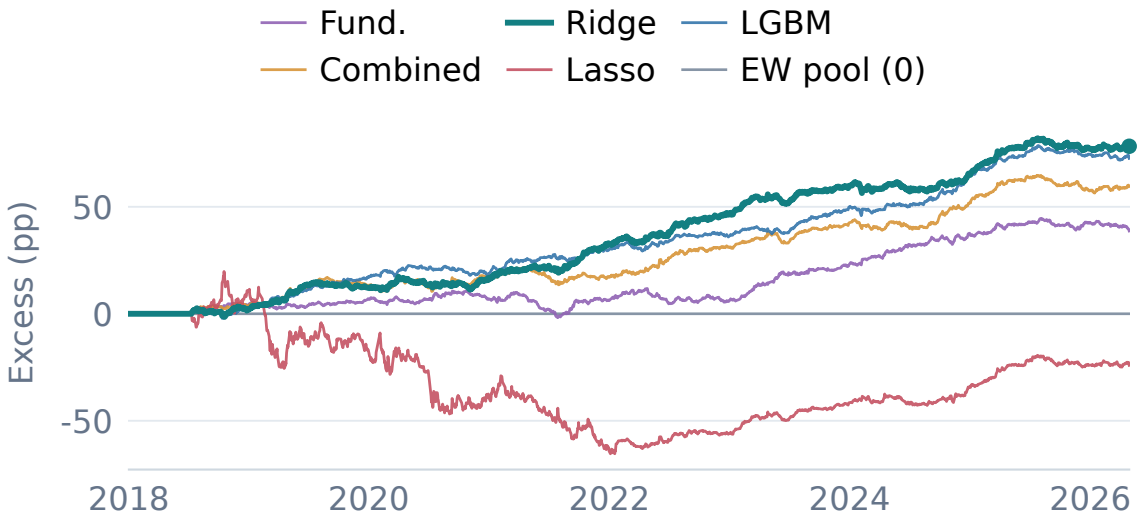


Figure 4. Cumulative daily gross excess versus the same-pool gross benchmark, 2 Jan 2018–29 Apr 2026.

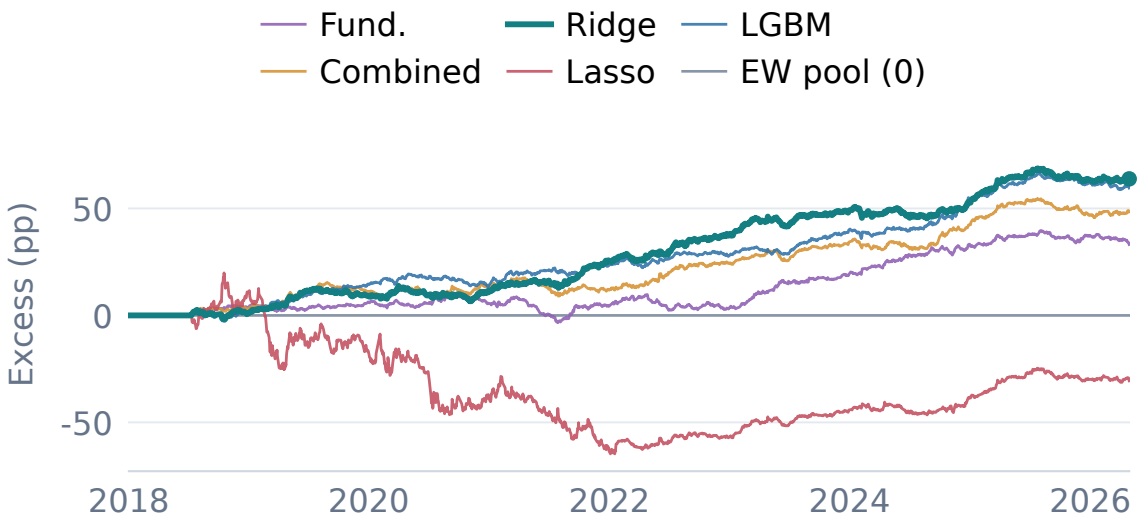


Figure 5. Cumulative daily net excess versus the same-pool net benchmark, including the matching execution costs.

Ridge accumulates +63.82 pp of daily net excess over the full history, including +52.02 pp in 2021–2025 and +0.56 pp in 2026 through 29 Apr 2026. The advantage is uneven: falling segments show intervals when the benchmark outperforms.

6. Drawdowns and complete-history risk

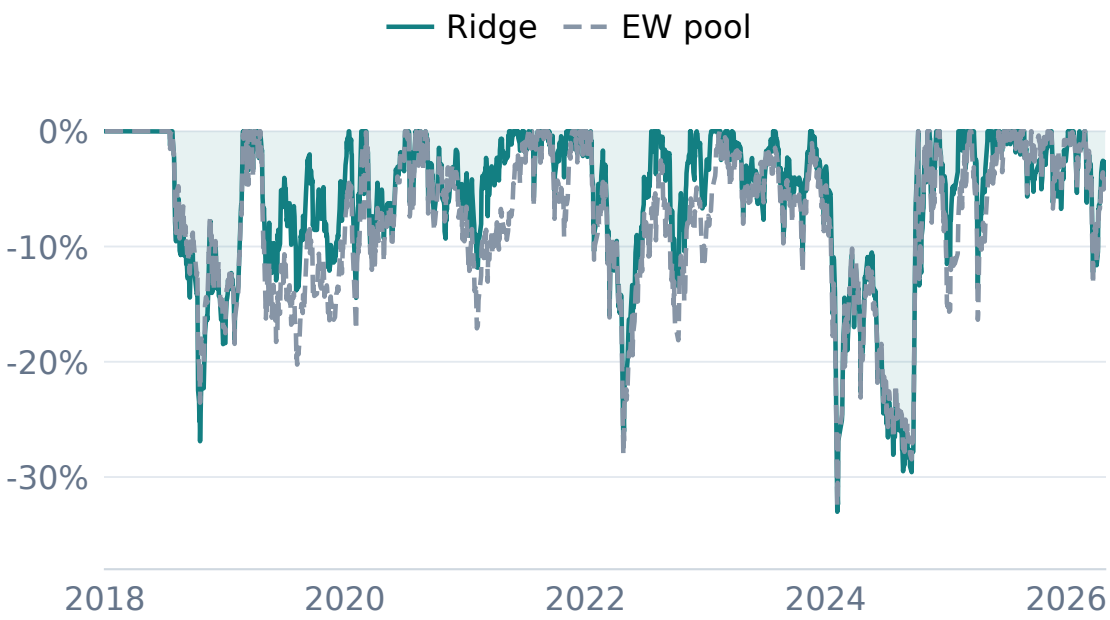


Figure 6. Net-wealth drawdowns relative to each portfolio’s prior peak. Negative values measure losses from that peak.

Portfolio	Net return	Net CAGR	Ann. vol.	Max. DD
Fundamental	258.8%	18.6%	24.8%	-30.3%
Combined	325.2%	21.3%	23.8%	-32.6%
Ridge	395.3%	23.8%	23.9%	-33.0%
Lasso	110.8%	10.5%	18.3%	-30.6%
LightGBM	360.4%	22.6%	25.5%	-33.7%
Benchmark	161.5%	13.7%	23.9%	-32.4%

Ridge’s worst drawdown is 33.0%, slightly deeper than the benchmark’s 32.4%. Its annualized volatility is 23.9%, compared with 23.9%. The historical return advantage does not remove market exposure, drawdowns or model risk. Annualization uses 252 market sessions; no risk-free return is assumed for Sharpe.

7. The return–risk trade-off in 2021–2025

The same 2021–2025 interval is used for every method. A higher point has a higher compounded annual growth rate; a point farther right has more annualized daily volatility.

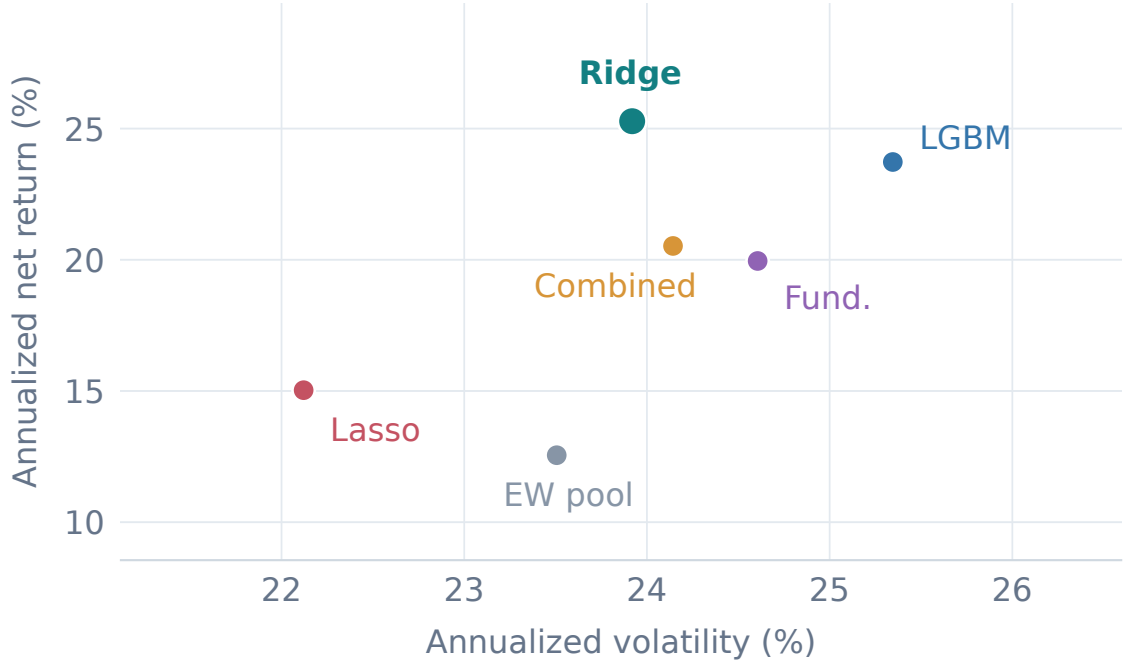


Figure 7. Net CAGR versus annualized daily volatility in 2021–2025, with identical 20-session target schedules and matched costs.

Ridge’s net CAGR is 25.3%, versus 12.6% for the benchmark, at volatilities of 23.9% and 23.5%. This historical advantage does not provide protection against large losses: the previous page shows comparable peak-to-trough risk.

LightGBM permits nonlinear interactions, but flexibility alone does not establish a better return–risk trade-off. The plotted history describes the adopted implementations rather than a universally superior model.

8. Performance across months and calendar years

Monthly excess sums daily return differences; annual returns compound wealth. Both expose variation hidden by the final cumulative return.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2018	0.0	0.0	0.0	0.0	0.0	0.0	+2.2	-1.2	-0.6	-0.1	+1.9	-1.2
2019	+1.7	+0.7	+1.1	+3.0	+1.4	+2.6	+1.0	-1.2	-0.6	-0.1	-0.5	-1.0
2020	+0.1	+0.1	+4.5	-1.7	-1.6	-1.1	-0.7	+0.8	+0.5	-2.3	+2.5	+1.0
2021	+2.7	+1.3	-1.0	+1.4	-0.3	+0.3	-2.8	+2.9	+3.1	+4.0	0.0	+2.5
2022	+2.1	+0.4	-2.0	+1.9	+1.1	+0.4	+3.8	+2.4	+0.1	+0.6	+0.3	+1.5
2023	+2.1	+3.1	+1.6	+0.6	-1.1	-2.0	+4.7	+0.2	+0.1	+0.2	+1.4	+0.4
2024	+0.5	-1.6	-0.7	+2.3	+0.2	-3.7	+0.7	-0.1	+2.5	+0.2	+0.8	+2.8
2025	+3.7	+2.4	+3.2	+2.5	+0.1	+1.5	+1.5	-0.8	-2.6	+0.1	-0.4	-0.9
2026*	+0.2	-0.3	+0.1	+0.5	—	—	—	—	—	—	—	—



Figure 8. Ridge monthly net excess in percentage points: the sum of daily strategy-minus-benchmark returns. Gray dashes denote unobserved months.

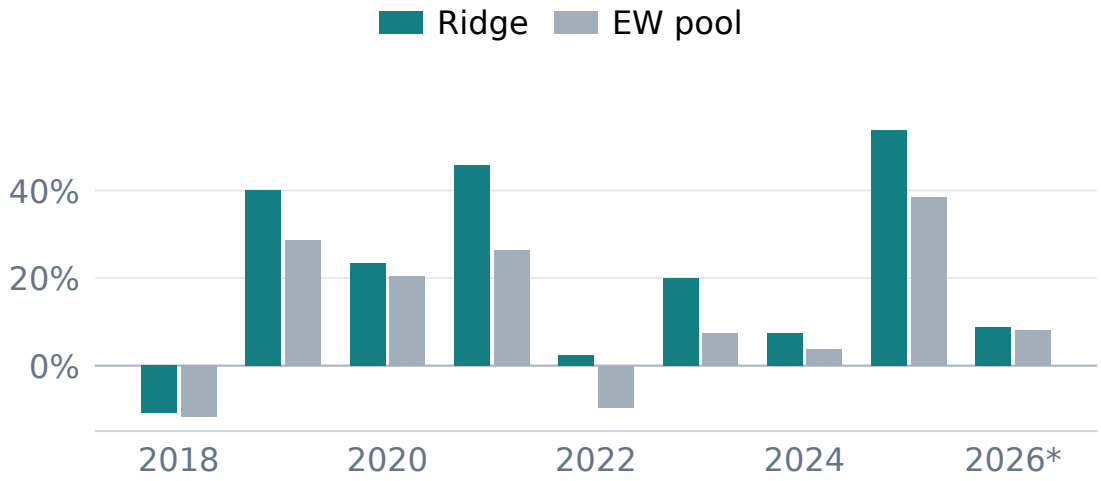


Figure 9. Compounded calendar-year net returns. The 2026 observations end on 29 Apr 2026; the final bar is a partial year.

The 2021–2025 Ridge net return is 195.7%, while 2026 through 29 Apr 2026 returns 8.80%. Negative monthly excess occurs within the profitable history. The partial 2026 bar is not directly comparable with full-year bars.

9. Turnover and the sources of trading costs

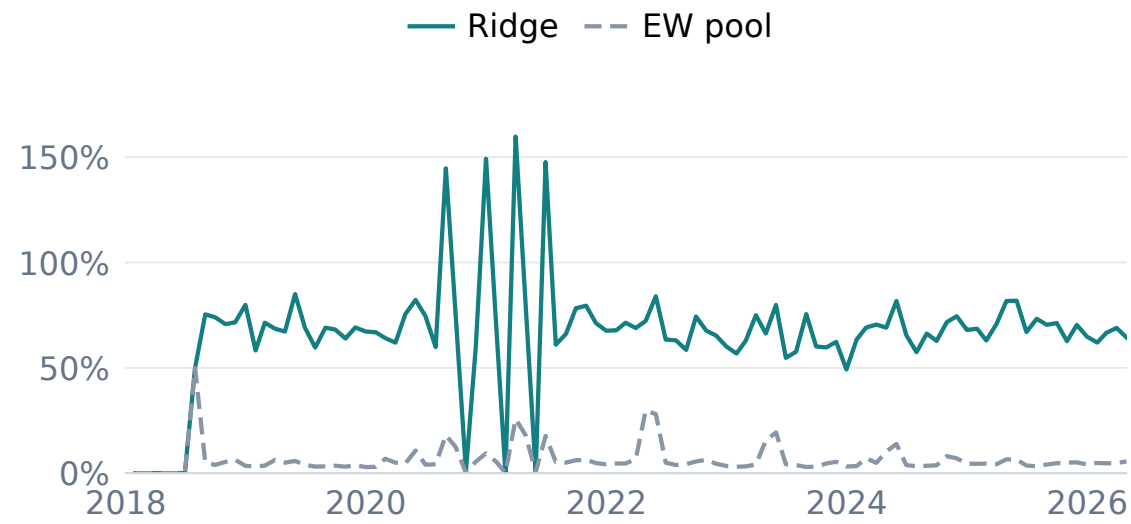


Figure 10. Monthly summed daily turnover for Ridge and the benchmark. Daily turnover is half the traded notional divided by opening wealth.

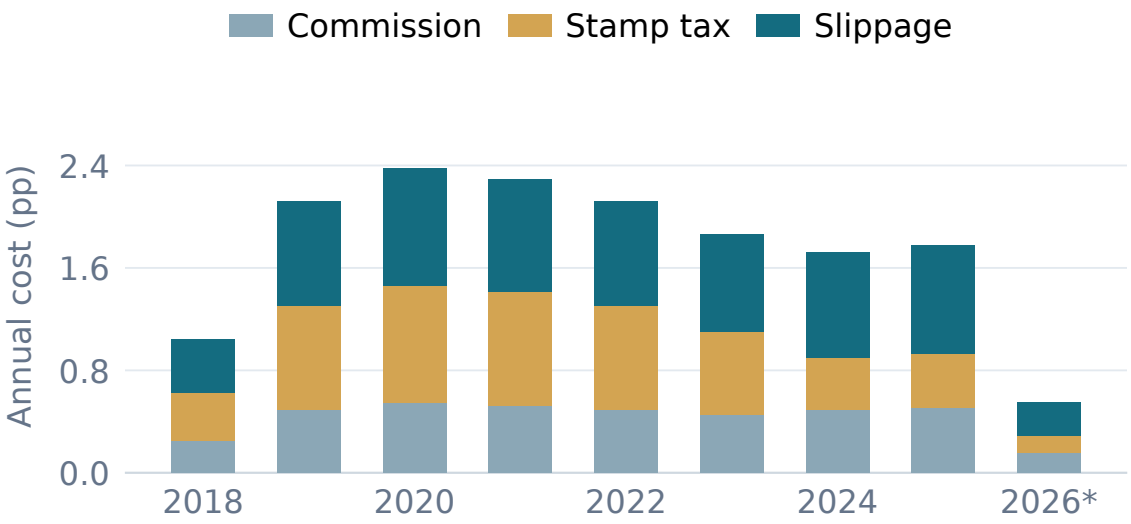


Figure 11. Calendar-year Ridge commission, stamp duty and slippage, scaled by each transaction day's pre-trade opening NAV and summed (pp). 2026 ends on 29 Apr; the 2023 tax change applies for part of that year.

Mean daily turnover is 3.46% for Ridge, versus 0.34% for the benchmark. Blocked-order retries can create trades between target refreshes. The cost bars are accounting charges, not the gross-minus-net terminal return difference.

10. Dependence among the adopted inputs

Daily cross-sectional rank correlations reveal which inputs carry overlapping information. They describe statistical dependence rather than causality or profitable trading after costs.

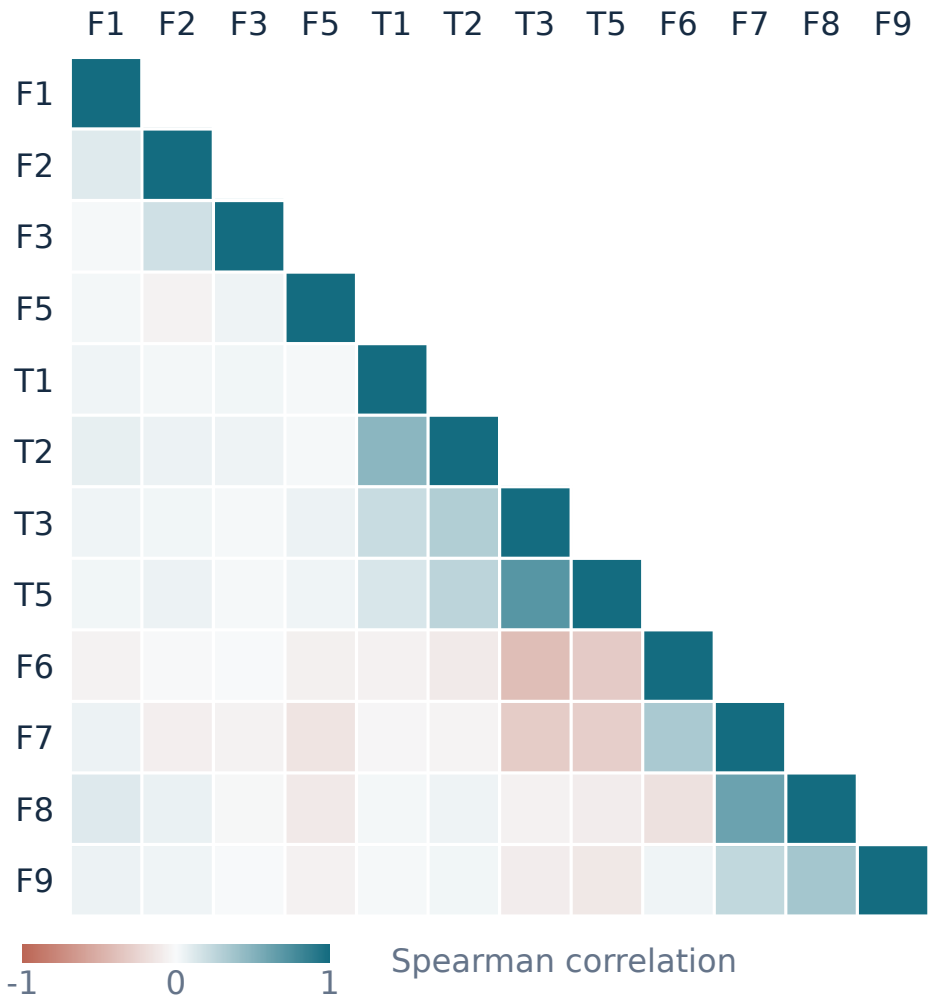


Figure 12. Adopted-input rank correlations: equal-date means of daily cross-sectional Spearman correlations in 2021–2025, sampled every fifth session in the common pool. The lower triangle uses color rather than a numerical label in every cell.

T3 and T5 have rank correlation +0.70, showing overlap between volatility and activity. Correlated inputs need not be removed mechanically, but they can repeat the same exposure. Ridge regularization controls coefficients; it does not create independent information.

11. Forecast rank correlation

Rank IC is the within-date Spearman correlation of a model score with the future 20-session opening return. A positive value means the full cross section is ordered in the intended direction on average; it does not directly describe the Top-100 trading portfolio.

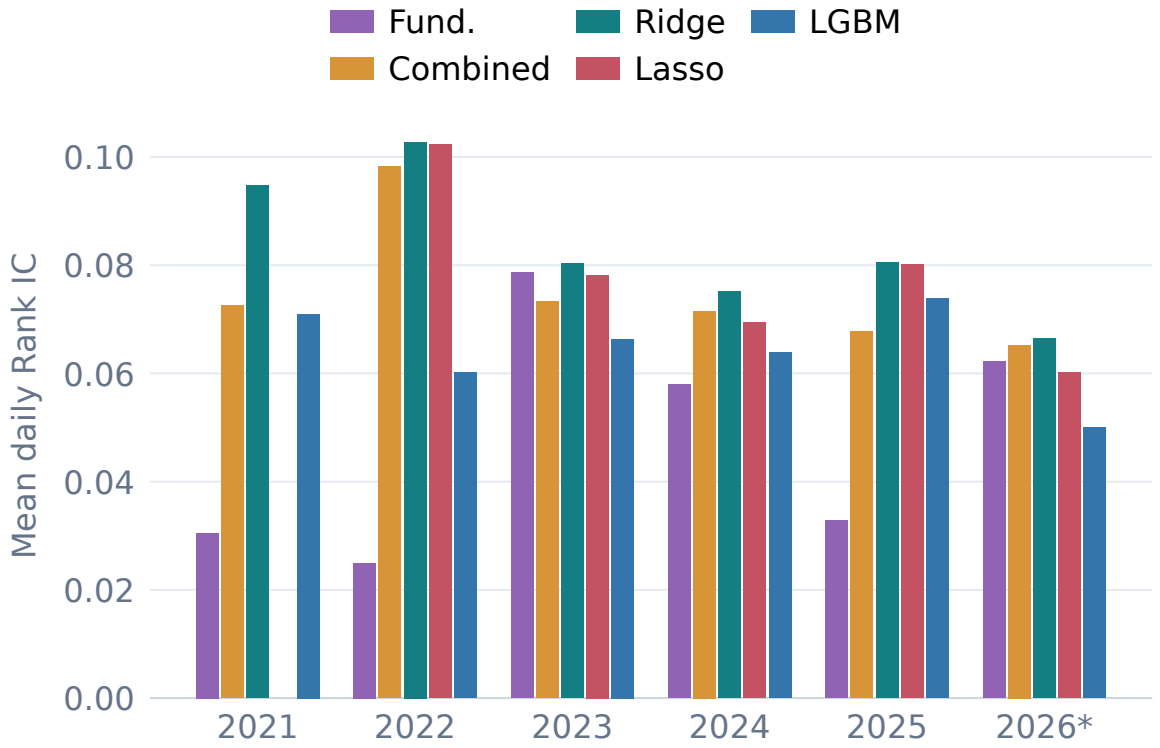


Figure 13. Annual mean daily Spearman rank IC of adopted scores against mature 20-session opening-return labels. The 2021–2025 values require label exits by 31 Dec 2025; 2026 exits end on 29 Apr 2026. Null-model years have no ranking and no IC bar.

Ridge’s 2026 mean IC is +0.0665, but a positive whole-pool IC can coexist with weak high-score selection. The 2026 average uses 55 mature signal dates through 30 Mar 2026. Overlapping labels make daily IC observations dependent; the chart is descriptive rather than an independent-day significance test. The next page examines the high-score groups directly.

12. Does the high-score group contain better returns?

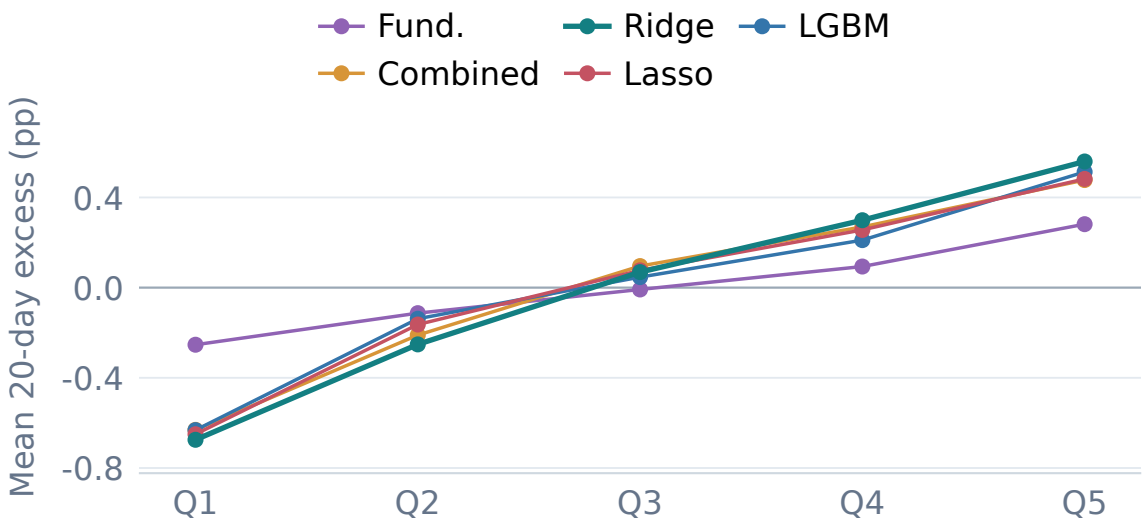


Figure 14. Mean future 20-session return excess by score quintile, 2021–2025. Q1 is the lowest score; Q5 is the highest.

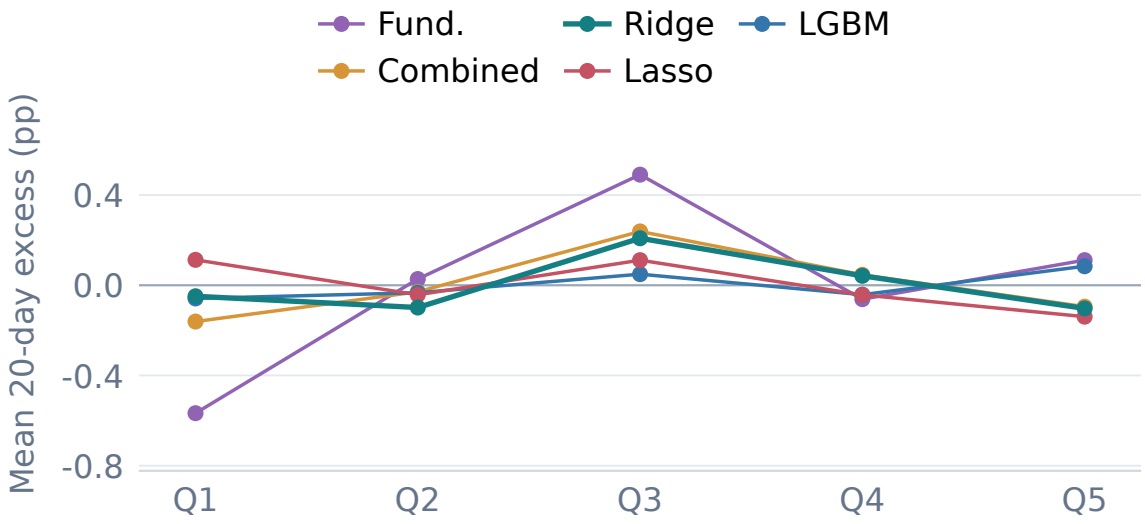


Figure 15. The same diagnostic for available 2026 signal dates through 30 Mar 2026. Later signals do not yet have mature labels.

These are future-label cohort averages, not portfolio returns: they ignore fees, opening execution limits, cash and delayed sales. The comparison uses near-equal score groups before removing unavailable labels; ties use fixed security order. Excess subtracts the daily labeled score-pool mean, and daily cohort means receive equal weight. Labels ending after 2025 are excluded from the 2021–2025 panel. Adjacent cohorts overlap and are not independent. Ridge's Q5-Q1 spread moves from +1.23 pp to -0.05 pp, weakening high-score separation in 2026.

13. Lasso: sparse coefficients and matched returns

Lasso minimizes squared error plus an L1 coefficient penalty. Unlike Ridge, it can set coefficients exactly to zero. Input ranks, annual history, industry/size processing, EMA20, Top-100 execution and costs match Ridge; only the regression and its historical regularization selection change. [5]

Interval	Lasso net	Ridge net	Lasso excess (pp)
2021–2025	96.10%	195.66%	+8.93
2026 partial	7.52%	8.80%	-0.65

Historical parameter selection

For each forecast year, the preceding calendar year supplies internal validation. Internal training labels must exit before validation begins; validation labels must exit before the forecast year. 15 alpha fractions of the training-only all-zero threshold are compared using equal-date mean validation squared error. The selected fraction is applied to the threshold recomputed on all mature history for the final annual fit. There is no random date split or forecast-year return selection.

Year	Alpha	Nonzero / 8	Mean Rank IC
2021	0.00114	0	— (no scores)
2022	0.000261	7	+0.1023
2023	1.29e-09	8	+0.0782
2024	0.000334	7	+0.0694
2025	1.15e-09	8	+0.0801
2026	0.000366	6	+0.0602

Annual final fits retain 0–8 of eight inputs. The 2026 nonzero inputs are F2, F5, T3, T5, T1, T2. A small coefficient is not evidence of economic irrelevance: correlated inputs can substitute for one another. Validation prediction loss, whole-pool Rank IC and fee-paying Top-100 returns measure different objectives. An all-zero fit emits no score: the next target refresh requests cash, subject to execution limits, rather than a security-order Top-100 basket.

Lasso does not improve on Ridge in either reported interval. The 2021 fit selects the null model; the weak-penalty 2023 and 2025 fits approach unpenalized regression. 3 of six fits select a final-grid boundary. The already reviewed 2026 history is not a new blind test. Full alpha, coefficient, maturity and convergence records accompany the results.

14. Limitations and conclusions

What the results support

Ridge and Lasso test different coefficient penalties under matched score processing and execution. The comparison records both return and sparsity; neither penalty guarantees stronger future returns or protection from drawdowns.

What remains uncertain

Historical specification choices can favor patterns that do not persist. The reviewed 2026 record is short. Overlapping labels are dependent; complete-input filtering changes the stock pool; vendor revisions remain uncertain even with announcement-date guards.

Daily bars omit queue priority and precise market impact. Corporate-action prices approximate reinvestment rather than exact dividend receipts and taxes. Residualizing scores does not enforce neutral actual holdings or lower risk.

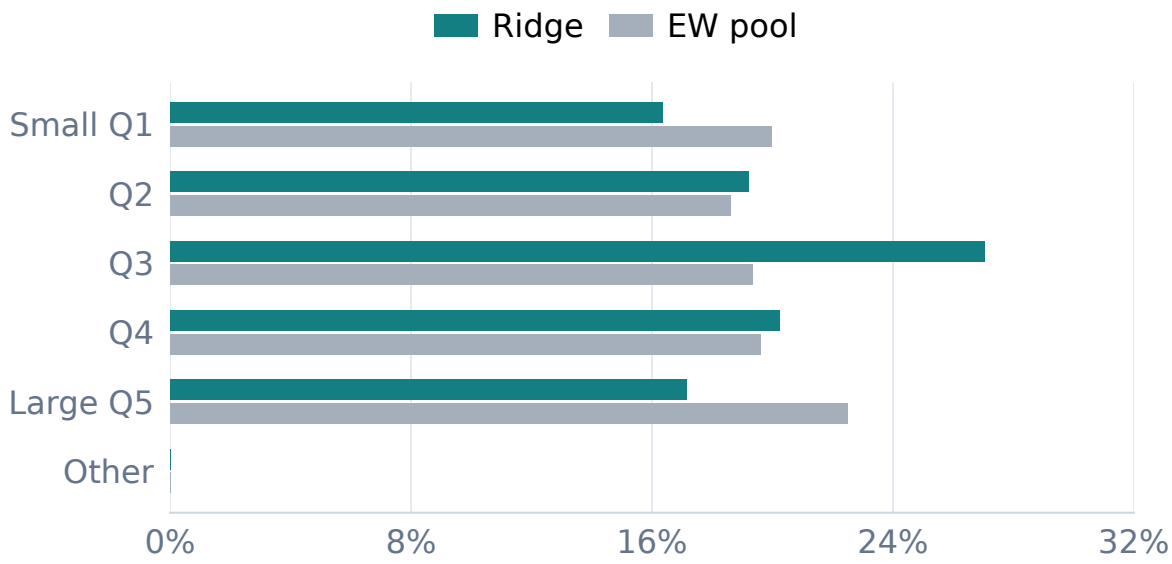


Figure 17. Actual net-NAV weights on 29 Apr 2026 by market-cap quintile (Q1 small, Q5 large), using same-day eligible-pool boundaries. Other combines cash, pending claims and unknown capitalization.

The historical net result warrants further study. Ridge’s 8.80% net return in 2026 through 29 Apr 2026, against 8.18% for the matched benchmark, is modest evidence over a partial year. A future prospective run with unchanged rules would provide a stronger check.

References and reproducibility

[1] Scikit-learn. *Ridge regression documentation.* Objective and L2-regularization parameter. Official API documentation.

[2] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). *LightGBM: A Highly Efficient Gradient Boosting Decision Tree.* Advances in Neural Information Processing Systems 30. Conference paper.

[3] Ministry of Finance, China (2008). *Securities transaction stamp duty changed to sell-side collection from 19 September 2008.* Official announcement.

[4] Ministry of Finance and State Taxation Administration, China (2023). *Announcement No. 39: Halving securities transaction stamp duty from 28 August 2023.* Official announcement. Commission and slippage are research assumptions.

[5] Scikit-learn. *Lasso regression documentation.* L1 objective, coefficient sparsity and convergence diagnostics. Official API documentation.

Reproduction from prepared external inputs

config.json records fixed annual factor selections, directions, model settings and trading assumptions. data_dictionary.md defines the prepared arrays, factor formulas, financial visibility guards and external corporate-action table. The calculation code retrains the fixed models and reconstructs all selected and raw-baseline portfolios from those inputs; no previous score or return file is required as a model input.

Generated external results contain the continuous daily ledger and metrics for the 20-session main implementation, daily comparison and daily doubled-slippage comparison. The raw baseline has its own matched 20-session ledger. Charts use those ledgers, while rank-IC and score-group figures use mature labels and adopted scores. Obtaining or rebuilding the supplier’s original financial snapshots is outside the prepared-input reproduction.